

NATIONAL JUNIOR COLLEGE
SH2 PRELIMINARY EXAM
Higher 1

CANDIDATE
NAME

SUBJECT
CLASS

REGISTRATION
NUMBER

CHEMISTRY

Paper 1 Multiple Choice

Additional Materials:

Multiple Choice Answer Sheet

Data Booklet

8872/01

Thursday 19 Sept 2013

50 minutes

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, subject class and registration number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **30** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This paper consists of **13** printed pages and **1** blank page.

Section A

For each question there are four possible answers **A**, **B**, **C**, and **D**. Choose the one you consider to be correct.

- 1** *Use of the Data Booklet is relevant to this question.*

A typical solid fertiliser for use with household plants and shrubs contains the elements N, P, and K in the ratio of 15 g : 30 g : 15 g per 100 g of fertiliser. The recommended usage of fertiliser is 14 g of fertiliser per 5 dm³ of water.

What is the concentration of nitrogen atoms in this solution?

- A** 0.03 mol dm⁻³
B 0.05 mol dm⁻³
C 0.42 mol dm⁻³
D 0.75 mol dm⁻³
- 2** In which species are the numbers of electrons and neutrons equal?
- A** $^{18}_8\text{O}^{2-}$
B $^{19}_9\text{F}$
C $^{23}_{11}\text{Na}^+$
D ^9_4Be
- 3** Which one of the following represents the configuration of the three electrons of highest energy for the ground state of an element in group III?
- A** 1s² 2s¹
B 1s¹ 2s¹ 2p¹
C 2s¹ 2p²
D 4s² 4p¹

- 4 A pure hydrocarbon is used in bottled gas for cooking and heating. When 10 cm^3 of the hydrocarbon is burned in 70 cm^3 of oxygen (an excess), the final gaseous mixture contains 30 cm^3 of carbon dioxide and 20 cm^3 of unreacted oxygen. All gaseous volumes were measured under identical conditions.

What is the formula of the hydrocarbon?

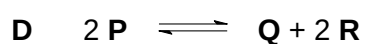
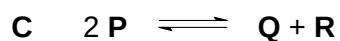
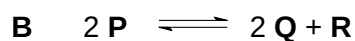
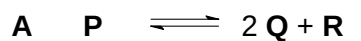
- A C_2H_6
B C_3H_6
C C_3H_8
D C_4H_{10}
- 5 The ions P^{3-} , S^{2-} and Cl^- have radii 0.212nm , 0.184nm and 0.181nm respectively. Which one of the following correctly explains the decreases in radius going from P^{3-} to Cl^- ?
- A Increases in both the total number of electrons and the nuclear charge
B An increase in the total number of electrons with the nuclear charge remaining constant
C A constant total number of electrons and an increase in the nuclear charge
D An decrease in the total number of electrons with the nuclear charge remaining constant
- 6 Which of the following molecules and ions does not have a bond angle of 120° around the central atom(s)?
- A AlCl_3
B PH_3
C CH_3^+
D N_2O_4
- 7 In which process are hydrogen bonds broken?
- A $2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{H}_2(\text{g}) + \text{O}_2(\text{g})$
B $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
C $\text{CH}_3\text{CH}_2\text{OH}(\text{l}) \rightarrow \text{CH}_3\text{CH}_2\text{OH}(\text{g})$
D $\text{CH}_3\text{COOH}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{CH}_3\text{COONa}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

- 8 Which of the following statements about lattice structures is **correct**?
- A The lattice energy of an ionic crystal tends to be large if the sizes of the two types of ions involved are small.
 - B The particles occupying lattice sites may be atoms or ions but not molecules.
 - C The forces holding the lattice together are so strong that the particles occupying lattice sites are held stationary.
 - D The lattice energy of an ionic crystal is always larger than that of a metal.
- 9 The gecko, a small lizard, can climb up a smooth glass window. The gecko has millions of microscopic hairs on its toes and each hair has thousands of pads at its tip. The result is that the molecules in the pads are extremely close to the glass surface on which the gecko is climbing.
What is the attraction between the gecko's toe pads and the glass surface?
- A co-ordinate bonds
 - B covalent bonds
 - C ionic bonds
 - D van der Waals' forces
- 10 The equation below represents the combination of gaseous atoms of non-metal **X** and of hydrogen to form gaseous **X₂H₆** molecules.
- $$2 \text{X(g)} + 6 \text{H(g)} \rightarrow \text{X}_2\text{H}_6\text{(g)} \quad \Delta H = -2775 \text{kJmol}^{-1}$$
- The bond energy of an **X–H** bond is 395kJmol^{-1} .
What is the bond dissociation energy of an **X–X** bond?
- A $-405.0 \text{ kJmol}^{-1}$
 - B $-202.5 \text{ kJmol}^{-1}$
 - C $+202.5 \text{ kJmol}^{-1}$
 - D $+405.0 \text{ kJmol}^{-1}$

- 11** Two moles of compound **P** were placed in a vessel. The compound **P** was partly decomposed by heating. A dynamic equilibrium between chemicals **P**, **Q** and **R** was established.

At equilibrium, x mol of **R** were present and the total number of moles present was $(2 + x)$.

What is the equation for this equilibrium?



- 12** The table gives the concentrations and pH values of the aqueous solutions of two compounds, **X** and **Y**. Either compound could be an acid or a base.

	X	Y
Concentration / mol dm ⁻³	10^{-4}	10^{-2}
pH	6	9

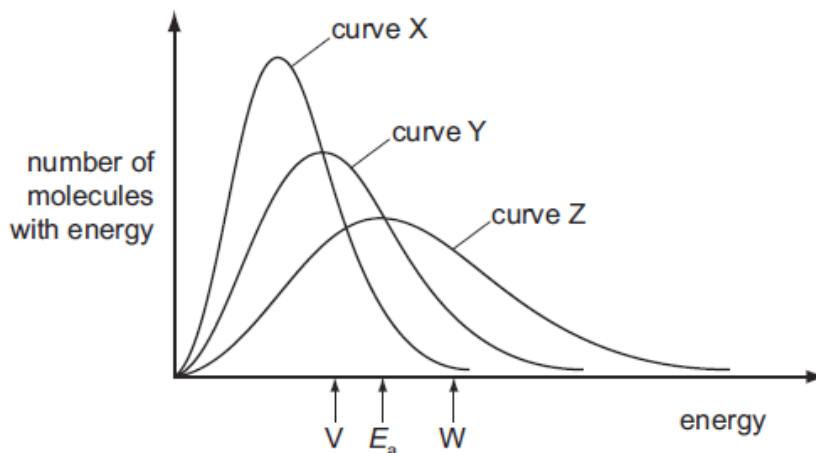
Student **P** concluded that **X** is a strong acid.

Student **Q** concluded that the extent of dissociation is lower in **X**(aq) than in **Y**(aq).

Which of the students are correct?

- A** both **P** and **Q**
B neither **P** nor **Q**
C **P** only
D **Q** only

- 13 The curve Y and the value E_a represent the distribution of energies of the molecules and the activation energy for an uncatalysed gaseous reaction.



What is a possible outcome if the reaction is catalysed?

- A The distribution of energies will be given by curve Y and the activation energy by value V.
- B The distribution of energies will be given by curve Y and the activation energy by value W.
- C The distribution of energies will be given by curve X and the activation energy by value W.
- D The distribution of energies will be given by curve Z and the activation energy by value V.

- 14 The following data is obtained during the first order thermal decomposition of



at constant volume and temperature.

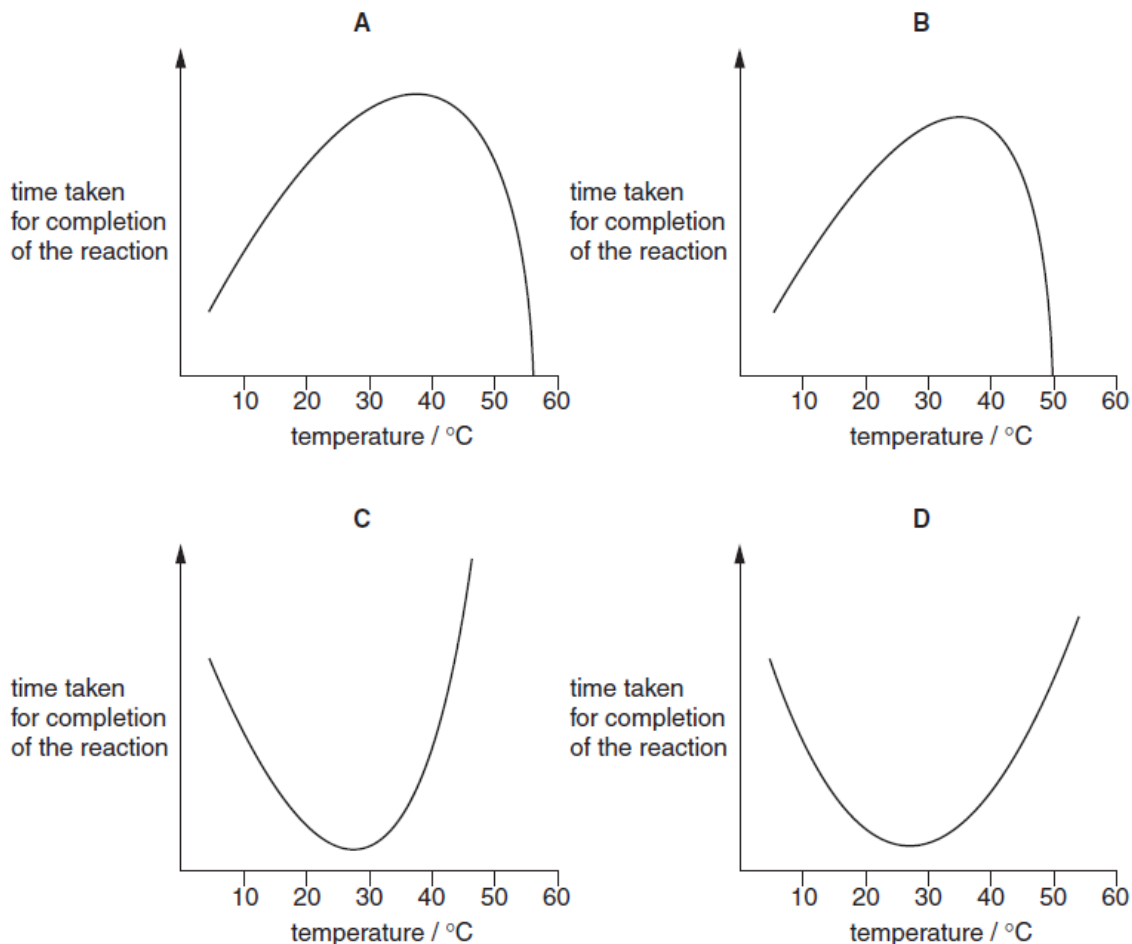
Time	Total pressure in Pascal
At the end of 10 minutes of reaction	600
After completion of reaction	400

What is the rate constant for this reaction in min^{-1} ?

- A cannot be determined due to insufficient data
- B 0.0346
- C 0.0693
- D 0.138

- 15 An enzyme is completely denatured at 50 °C. A fixed concentration of this enzyme is added to a fixed concentration of its substrate. The time taken for completion of the reaction is measured at different temperatures.

Which graph shows the results?



- 16 The information relates to element Z.

- Z is in Period 3 of the Periodic Table.
- Z has a lower electrical conductivity than Mg.
- A Z atom has a half-filled subshell in its ground state.
- Z forms an acidic oxide on exposure to air.

What is Z?

- A Na
B Si
C P
D Cl

- 17 One of the reactions taking place in a catalytic converter in a car exhaust system is between nitrogen oxide and octane (unburned petrol). The products of this reaction are non-toxic.

Which is the correct equation for the reaction?

- A $\text{C}_8\text{H}_{16} + 16 \text{NO} \rightarrow 8 \text{CO} + 8 \text{N}_2 + 8 \text{H}_2\text{O}$
- B $\text{C}_8\text{H}_{16} + 24 \text{NO} \rightarrow 8 \text{CO}_2 + 12 \text{N}_2 + 8 \text{H}_2\text{O}$
- C $\text{C}_8\text{H}_{18} + 17 \text{NO} \rightarrow 8 \text{CO} + 8 \frac{1}{2} \text{N}_2 + 9 \text{H}_2\text{O}$
- D $\text{C}_8\text{H}_{18} + 25 \text{NO} \rightarrow 8 \text{CO}_2 + 12 \frac{1}{2} \text{N}_2 + 9 \text{H}_2\text{O}$
- 18 The products obtained by cracking an alkane, **M**, are methane, ethene and propene. The mole fraction of ethene in the products is 0.5.

What is the identity of **M**?

- A C_6H_{14}
- B C_8H_{18}
- C C_9H_{20}
- D $\text{C}_{11}\text{H}_{24}$
- 19 What is the total number of structural isomers for an organic compound, $\text{C}_4\text{H}_9\text{F}$?
- A 3
- B 4
- C 5
- D 6

- 20 *Use of the Data Booklet is relevant to this question.*

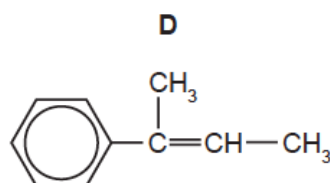
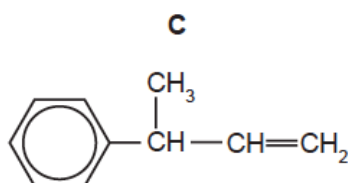
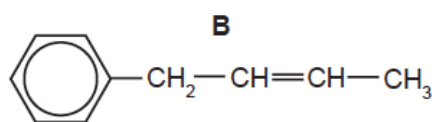
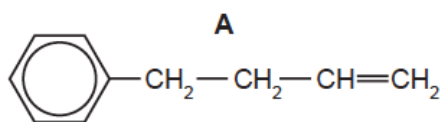
In which reaction is the relative molecular mass of the organic product the largest?

- A bromoethane + aqueous sodium hydroxide
- B bromoethane + alcoholic sodium hydroxide
- C ethane + bromine
- D ethanol + phosphorus pentachloride

21 Compound X

- has the molecular formula $C_{10}H_{14}O$;
- is unreactive towards mild oxidising agents.

What is the structure of the compound formed by dehydration of X?

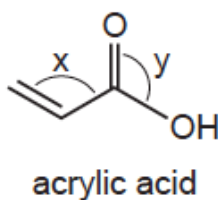


- 22 An organic substance **S** reacts with sodium to give hydrogen, but does not react with 2,4-dinitrophenylhydrazine and does not react with ethanol in the presence of concentrated sulfuric acid.

What can be deduced from this information?

- A** **S** is a carboxylic acid.
B **S** is a ketone.
C **S** is an alcohol.
D **S** is an alkane.

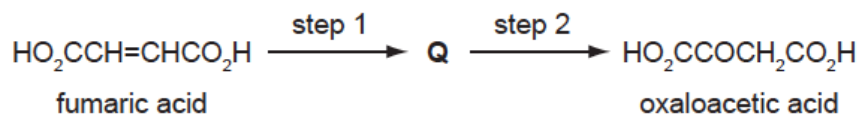
- 23 Acrylic acid is produced from propene, a gaseous product of oil refineries.



Which statement about acrylic acid is **not** correct?

- A** Both bond angles x and y are approximately 120° .
B It decolourises aqueous bromine.
C It gives an orange precipitate with 2,4-dinitrophenylhydrazine reagent.
D It reacts with an alcohol to give an ester.

- 24 Fumaric acid can be converted in to oxaloacetic acid by a two-step process involving the intermediate **Q**.

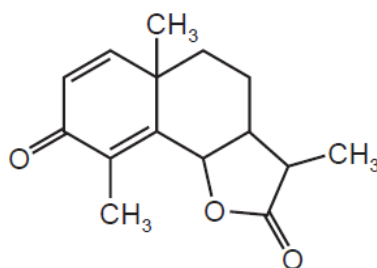


Each of these steps can be achieved in the laboratory by a single reagent.

What could be the intermediate **Q** and the reagent for step 2?

	Q	Reagent for step 2
A	$\text{HO}_2\text{CCH}(\text{OH})\text{CH}_2\text{COOH}$	Warm acidified $\text{K}_2\text{Cr}_2\text{O}_7$
B	$\text{HO}_2\text{CCHBrCH}(\text{OH})\text{COOH}$	Warm $\text{NaOH}(\text{aq})$
C	$\text{HO}_2\text{CCH}(\text{OH})\text{CH}_2\text{COOH}$	Fehling's solution
D	$\text{HO}_2\text{CCHBrCH}_2\text{COOH}$	Warm acidified KMnO_4

- 25 Santonin is a drug that was once widely used to expel parasitic worms from the body.



Santonin

When santonin is first treated with warm dilute H_2SO_4 , followed by cold acidified KMnO_4 , **X** is obtained.

How many atoms of hydrogen in each molecule of **X** can be displaced with sodium metal?

- A** 2 **B** 4 **C** 5 **D** 6

Section B

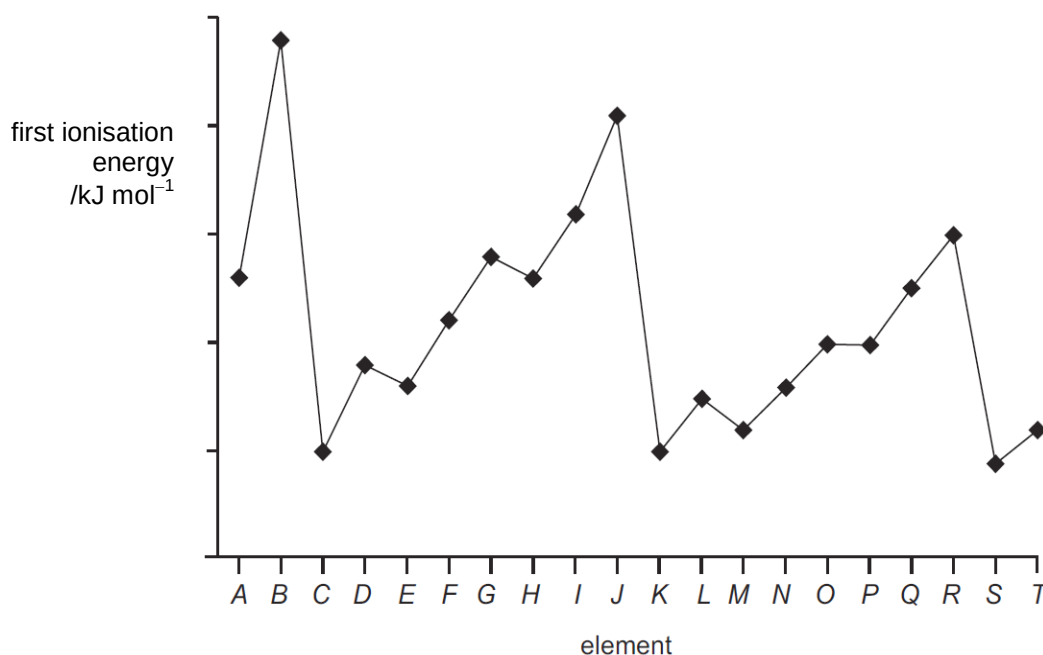
For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct. Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2, and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

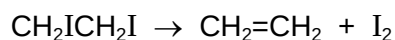
- 26** The first ionisation energies of twenty successive elements in the Periodic Table are represented in the graph.
The letters given are not the normal symbols for these elements.



Which statements about the elements and their compounds in this graph are correct?

- 1** Elements **B**, **J** and **R** are in Group 0 of the Periodic Table.
- 2** The second ionisation energy of **K** is larger than the second ionisation energy of **L**.
- 3** Oxide and hydroxide of element **M** are amphoteric.

- 27 When light is shone for a fixed period into a solution of $\text{CH}_2\text{ICH}_2\text{I}$ and I_2 in tetrachloromethane at 100°C , the following reaction occurs:

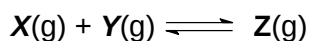


experiment	Relative initial concentration		Relative light intensity	Relative initial rate of formation of I_2
	$\text{C}_2\text{H}_4\text{I}_2$	I_2		
1	2	1	4	2
2	2	1	1	1
3	1	1	4	1
4	1	2	4	1

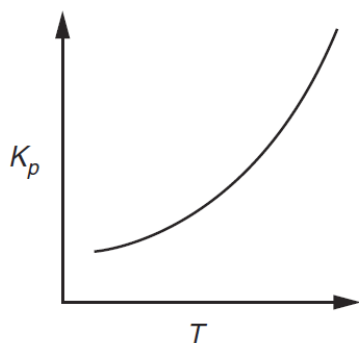
What conclusions about the reaction can be drawn from the data in the table?

- 1 The rate of formation of iodine is proportional to initial concentration of $\text{C}_2\text{H}_4\text{I}_2$.
- 2 The rate of formation of iodine is proportional to $\sqrt{(\text{light intensity})}$.
- 3 Iodine is not involved in the rate determining step.

- 28 The equilibrium constant K_p for the reaction



varies with temperature as shown in the diagram below.



Which conclusions can be drawn from this information?

- 1 The equilibrium mixture contains a greater proportion of **Z** at higher temperatures.
- 2 The equilibrium mixture contains a greater proportion of **Z** at higher pressures.
- 3 The activation energy for the forward reaction is smaller than the activation energy for the backward reaction.

The responses **A** to **D** should be selected on the basis of

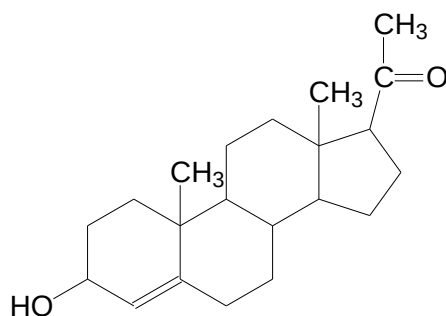
A	B	C	D
1, 2, and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

29 Which pairs of reagents will react together in a redox reaction?

- 1** $\text{CH}_4 + \text{Cl}_2$
- 2** $\text{CH}_3\text{CH}_2\text{CHO} + \text{Fehling's reagent}$
- 3** $\text{CH}_3\text{CH}_2\text{COCH}_3 + \text{Tollens' reagent}$

30 The compound shown is a hormone produced during pregnancy to suppress ovulation.



Which statements are correct?

- 1** When reacted with alkaline aqueous iodine, a yellow precipitate is observed.
- 2** When reacted with LiAlH_4 , the product formed has 3 carbon atoms converted from sp^2 hybridised to sp^3 hybridised.
- 3** When reacted with an excess of hot acidified $\text{K}_2\text{Cr}_2\text{O}_7$, the product formed does not react with aqueous with aqueous Br_2 .

– End of Paper–

SH2 Prelim Paper 1 Answers

1	A	11	B	21	D
2	A	12	B	22	C
3	D	13	A	23	C
4	C	14	C	24	A
5	C	15	C	25	D
6	B	16	C	26	A
7	C	17	D	27	A
8	A	18	B	28	B
9	D	19	B	29	B
10	D	20	C	30	D